

Philosophy of Science
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Lecture 9 Rationality

Where in the course are we?

IV. METHODOLOGICAL TOOLBOX

7. Analogies and models
8. Quantitative and qualitative approaches

V. RATIONALITY AND CULTURE

9. Rationality ← *Today*
10. Interpretation and understanding
11. Absolutism and relativism
12. Review and question period

2

Motivation of this lecture

- . International studies often involves...
 - . Modelling and evaluating agents' behaviour in the light of standards of rationality
 - . Examples: economic actors, political leaders, voters, other decision makers
- . An understanding of the concept of rationality will help you...
 - . Make sense of the behaviour of such agents
 - . As well as reflect on your own standards of reasoning and decision making!

3

Plan of this lecture

- . What is rationality?
- . Two applications of models of rationality
 - . Descriptive and normative
- . Rational choice theory
 - . Criticisms of this approach
- . Alternative approaches to rationality
 - . Bounded rationality theory
 - . Theories of emotion and rationality

4

Reading for this lecture

- Herbert A. Simon, "Rationality in Psychology and Economics"
 - *Journal of Business* 59 (1986), pp. S209–S224
 - Available on Brightspace

5

What is rationality?

- Rationality: a property of an action
 - The property of being optimal under a set of constraints
- Relevant actions include especially:
 - Inferences, or the drawing of conclusions from premises and evidence—"theoretical rationality"
 - Decisions, such as a choice of option *A* over option *B*—"practical rationality"
- Further details: models of rationality

6

Models of rationality: two uses

- Descriptive use
 - How do agents in fact reason and decide?
 - E.g. economic actors, political leaders
 - Aim: model, explain, understand, and predict behaviour
- Normative use
 - How should agents reason and decide?
 - Aim: guide, assess, and criticise behaviour

7

Three approaches to rationality

- Each offers a characterisation and account of rationality
 - For descriptive and normative uses
- Rational choice theory (RCT)
 - Most influential twentieth-century model of rationality
- Two responses to RCT
 - Bounded rationality theory
 - Emotion-inclusive accounts of rationality

8

RCT: history 1

One influential tradition: rational deliberation concerns means to ends

[W]e deliberate, not about ends, but about what forwards those ends. For a doctor does not deliberate about whether he'll make his patients healthy, nor a public speaker about whether he'll persuade his audience, nor a political expert about whether he'll bring about good government—and neither do any of the others deliberate about the end, but rather they take the end for granted and examine how and by what means it will come about....

—Aristotle, *Nicomachean Ethics* (350 BCE), Book III, 3
Trans. by Christopher Rowe, 2002

9

RCT: history 2

- Philosophy
 - Jeremy Bentham (1748–1832) formulated utilitarianism: promoting “the greatest happiness of the greatest number”
- Mathematics
 - John von Neumann (1903–1957) and Oskar Morgenstern (1902–1977) formalised utility theory

10

RCT: history 3

- RCT is now an influential paradigm in economics
 - Especially microeconomics
- Criminology
 - Gary Becker (1930–2014) applied rational choice models to crime and discrimination
 - Presented as an amoral analysis
- Further domains
 - Political science, sociology, other social sciences

11

RCT: principles

- Framework of RCT
 - Given an aim, an agent is modelled as choosing the action that best fulfils that aim
 - by providing maximum net benefit
 - Net benefit = benefit – cost
- Idealisations: the agent has...
 - clear and consistent preferences
 - accurate beliefs about possible outcomes
 - enough computational capacity to select the best option

12

RCT: utility function

- RCT represents the agent's preferences with a utility function, $U = U(x, y, z, \dots)$
 - Assigns a numerical value of U to each possible alternative facing the agent
 - Higher values of U : more preferred alternatives
- Example:
 - Choice between alternative houses to buy
 - Many different variables: cost, location, size, features, condition...
 - $U_{\text{House 1}} > U_{\text{House 2}}$

13

RCT: problem of preferences 1

- RCT takes agents' preferences as given
 - It is silent on how agents form them
 - Given a certain set of preferences, RCT models an agent's deliberation
- RCT often treats preferences as "revealed" by choices...
 - But if preferences are inferred from the very behaviour they are meant to explain, the explanation risks becoming circular

14

RCT: problem of preferences 2

- Risk: if we are free to attribute whatever preferences would make an action optimal...
 - then almost any action can be reconstructed as rational
- Consequence of an instrumental notion of rationality

15

RCT: applications

- Prediction of consumer behaviour
 - At the outset, we do not know how consumers decide purchases
 - However, we can observe the purchases that consumers actually make
 - RCT assumes that these purchases are the optimal choices given certain preferences
- We reconstruct these preferences...
 - and use this theoretical model to predict future purchase patterns

16

RCT: discussion

- What does RCT offer in practice?
 - Framework for reconstructing agent's ideal motives
 - Given the agent's observed behaviour, what preferences could have motivated it?
 - Assumption: agent flawlessly translates motives into action
- RCT embodies an ideal reasoner
 - Perfect calculating agent
 - Example of idealised model—see lecture 7

17

RCT: criticisms 1

- RCT lacks resources to evaluate goals
 - Therefore, it tends to infer that an agent is motivated by the goals that would make their observed actions rational
 - Irrespective of how odd or unwise those acts seem
- RCT lacks a strong concept of non-rational or irrational action
 - Problem: does a theory that can explain everything have determinate content?

18

RCT: criticisms 2

- Humans are not perfect calculators
 - RCT may fail to capture actual human decision making
- Social and feminist critiques of RCT
 - RCT models individualistic choices motivated by self-interest
 - It can underplay care, dependency, socialisation, power, institutions, and collective agency

19

Further developments

- Extension of RCT to agents cooperating or competing with one another
 - Game theory—John Nash (1928–2015)
 - Examples: prisoner's dilemma, arms-race strategy and deterrence in the Cold War
- Concern for human cognitive limitations
 - Bounded rationality theory
- Embodied approaches to rationality
 - Accounts of emotion in reasoning and decision making

20

From RCT to bounded rationality

- In practice, humans are limited in...
 - information, cognitive powers, time
- Actual human decision making may be less idealised than RCT suggests
 - Rough and ready procedure
- Agents do not demand ideal outcomes
 - They want to do well enough: to satisfice ("satisfy" + "suffice"), not to optimise
 - Rejection of perfectionism

21

What is bounded rationality?

- Bounded rationality is...
 - a restricted form of rationality...
 - that is cheap and easy to implement...
 - and that gives results that are good enough
- Concept of "heuristic"
 - Context-dependent rule of thumb for reasoning and decision making
- Twin descriptive and normative claims
 - People actually use these heuristics
 - Their use is justified: they give good results

22

Bounded rationality: history

- Economics
 - Herbert A. Simon (1916–2001) coined the terms "bounded rationality", "to satisfice"
 - *Administrative Behavior* (1947): managers look for solutions that are satisfactory rather than ideal
- Psychology
 - Gerd Gigerenzer (1947–) itemised "fast and frugal" heuristics
 - Emphasised the value of partial ignorance

23

Examples of heuristic

- Task: choosing between alternatives
 - Recognition heuristic—exploits a lack of knowledge
 - Take-the-best heuristic—deliberately ignores information
- Descriptive and normative claims
 - This is how people actually reason...
 - and there are sound reasons for doing so

24

Recognition heuristic

- If you recognise only one of the alternatives, infer that that has the higher value of the variable of interest
 - Justification: a stronger signal is more likely to have reached you
 - Partial ignorance is here an advantage
- Experiment:
 - Which city is larger, Hanover or Bielefeld?
 - Who won Wimbledon in 2012, Agnieszka Radwańska or Serena Williams?

25

Take-the-best heuristic

- Given alternatives and a ranking of criteria, make a choice on the highest-ranking criterion that discriminates between the alternatives
 - Ignore all subsequent criteria
 - Deliberately ignores information
- Example: choosing a house
 - List features in order of importance to you
 - Choose the house with the highest score on the first feature with unequal scores

26

Bounded rationality: application

- Prediction of consumer behaviour
 - Consumers show certain patterns of purchases
 - Bounded rationality theory assumes that these patterns are governed by rules of thumb
- We reconstruct these rules of thumb...
 - and use this theoretical model to predict future behaviour

27

Bounded rationality: critique

- Some actual human decision making may be more rigorous than this
 - It may involve more calculation than Gigerenzer's simple rules
- Puzzle: how does an agent rationally choose which heuristic to use?
 - When is it appropriate to implement a particular heuristic, and when are there reasons to reject it?

28

Rationality and emotion

- Enduring polarity in Western culture
 - Emotion as antagonistic to rationality
 - Best decision makers are disembodied calculators—rational choice theory
- Cognitivism about emotion, since 1980
 - Rationality involves emotional responses to situations
 - Reopens discussion about aptness of goals
 - Emphasises the emotionally connected self
 - Example: Antonio Damasio (1944–)

29

Emotion in practical reasoning

- Two functions of the emotion system
 - Motivational function: emotion helps set goals, values, preferences
 - Salience function: emotion help identify what matters in a situation
- Important especially in cases in which...
 - Goals conflict, as in dilemmas
 - Information is incomplete, inconsistent, ambiguous or too abundant to review

30

Comparing the three accounts

- Case: a government considers imposing sanctions
 - RCT: which policy maximises expected national interest?
Bounded rationality: decision makers use limited information, rules of thumb, political cues, institutional routines
Emotion-inclusive accounts: fear, anger, trust, humiliation or solidarity shape what appears salient and what goals seem appropriate

31

Recapitulation

- Rationality is...
 - the property of an action of being optimal under a set of constraints
- Two uses of models of rationality
 - Descriptive and normative: modelling and guiding behaviour
- Three models of rationality
 - Rational choice theory, bounded rationality theory, theories of emotion in rationality

32